WORKED SOLUTION

STEP II Specimen paper, Question 10

Pure • Logic Puzzles

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q10

The key idea

Everything rests on one translation: a speaker is a toad exactly when what they say is true. So for each character we ask "could they be a toad?" and "could they be a frog?", and throw out any choice that makes their statement have the wrong truth value. Watch especially for statements about the speaker themselves: some of them can never be said by one kind of creature at all, and that is usually the way in.

Solution

Throughout, "toad" means a truth-teller and "frog" means a liar, so

$$\text{speaker is a toad} \iff \text{their statement is true.}$$

In each scene we find every assignment consistent with this rule, and check that exactly one survives.

Part (i)

Suppose A were a toad. Then A's statement would be true, so in particular A would be a frog, which is a contradiction. Hence A is a frog.

So A's statement is false: it is *not* the case that both A and B are frogs. Since A is a frog, the only way this can fail is for B to be a toad.

Check: with A a frog and B a toad, "both of us are frogs" is false, as a frog's statement must be. So A **is a frog and** B **is a toad**.

Part (ii)

This is the mirror image of part (i). Suppose C were a frog. Then "at least one of D and myself is a frog" would be true (because of C alone), but a frog cannot say something true. Hence C is a toad.

So C's statement is true: at least one of C, D is a frog. Since C is a toad, D must be the frog.

Check: with C a toad and D a frog, the statement is true, as a toad's must be. So C **is a toad and** D **is a frog**.

Part (iii)

Write S for E's statement, "G and H are both toads". Then G asserts that S is true, and H asserts that S is false.

Ask yourself We know nothing about anyone yet. Where can we possibly start?

Look at G and H together rather than at E. Their two statements are exact opposites, so exactly one of them is true. Hence exactly one of G, H is a toad and the other is a frog, whatever S turns out to be.

That settles S: since one of G, H is a frog, they are not both toads, so S is false. Therefore:

- E said something false, so E is a frog;
- G claimed S is true, which is false, so G is a frog;
- H claimed S is false, which is true, so H is a toad.

Check: G is a frog, so "G and H are both toads" is indeed false, consistent with every verdict. So E **and** G **are frogs and** H **is a toad**.

Part (iv)

Here "us" means the three of I, J, K.

I is a frog. If I were a toad, "all of us are frogs" would be true and so I would be a frog, a contradiction. So I is a frog, and I's statement is false: not all three are frogs, so

$$\text{at least one of } J,\ K \text{ is a toad.} \qquad (*)$$

J is a toad. Suppose instead J were a frog. Then by $(*)$, K is a toad, and the creatures are I frog, J frog, K toad: exactly one toad. That makes J's statement true, which a frog cannot say. So J is a toad.

K is a frog. Since J is a toad, J's statement is true: there is exactly one toad among the three. That toad is J, so K is a frog.

Ask yourself Does $(*)$ still hold now that K is a frog?

Yes: $(*)$ needs only one of J, K to be a toad, and J is. So the full check is: I's "all of us are frogs" is false (J is a toad), and J's "exactly one of us is a toad" is true (only J). Both statements have the right truth value, so I **and** K **are frogs and** J **is a toad**.

Checks

Each scene has only 2^2 or 2^3 possible assignments, so a complete check is a short truth table: for each assignment, test whether every speaker's statement is true exactly when that speaker is a toad. We ran this exhaustive search for all four scenes, and in each one exactly one assignment survives, the one found above. (The uniqueness matters: the question asks us to *decide*, so we need to rule the others out, not just find one that works.)

Common slips

- Finding a consistent assignment and stopping. You must also show that no other assignment works, which is what the "suppose it were…" contradictions do.
- In part (iii), treating G's "this is true" as a statement about G rather than about E's claim.
- In part (iv), forgetting that K, who says nothing, is still included in "all of us" and "exactly one of us".

Beyond STEP

Puzzles like these are exercises in propositional logic: each creature is a Boolean variable t, each statement a formula P, and every speaker imposes the constraint $t \iff P$. Deciding whether such a system of constraints has a solution is the *satisfiability* problem, which is central to theoretical computer science, and part (i)'s self-defeating statement is a gentle cousin of the liar paradox. Raymond Smullyan's *What Is the Name of This Book?* is a delightful collection of knights-and-knaves puzzles that builds up to Gödel's incompleteness theorem.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.